MUHAMMAD SHAYAN AYUB

Cybersecurity Analyst | Application Security | Backend Engineering

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PROFESSIONAL SUMMARY

Computer Science graduate and Developer with practical hands-on experience in cybersecurity investigations, network traffic analysis, and secure application development. Proven background in PCAP-based malware forensic analysis (DanaBot investigation using Wireshark and NetworkMiner), HTTP security header evaluation, OWASP Top 10 vulnerability remediation, and ethical hacking fundamentals (Top 25% ranking across 22+ TryHackMe rooms). Complemented by professional full-stack development experience building authenticated backend services and APIs using Python, FastAPI, PostgreSQL, and Docker. Seeking entry-level cybersecurity, SOC analyst, or application security roles.

CYBERSECURITY & TECHNICAL SKILLS

Cybersecurity & Network Analysis: OWASP Top 10 Mitigation, HTTP Security Headers, Wireshark, PCAP Network Forensics, NetworkMiner, Nmap, Malware Traffic Analysis, C2 Identification, Threat Intelligence (VirusTotal, ANY.RUN)

Application Security & Backend: Secure API Design, JWT Authentication, Password Hashing, Role-Based Access Control (RBAC), FastAPI, Python, WebSockets

Cloud Security & Infrastructure: Microsoft Azure Security Concepts, Cloud Governance, Monitoring & Telemetry, Docker Container Security, Linux (Debian, Kali)

Databases & Data Integrity: PostgreSQL, MySQL, SQLite, Relational Modeling, pgAdmin

Languages & Scripting: Python, JavaScript, C++, SQL, Bash Basics, HTML5, CSS3, PHP

DevOps & Development Tools: Git, GitHub, Docker, Postman, VS Code

CYBERSECURITY EXPERIENCE & PRACTICAL TRAINING

CyberDefenders — DanaBot Digital Forensics Lab

April 2025 | Verification Record

- Executed an in-depth PCAP-based digital forensics investigation into a suspected DanaBot malware infection.
- Analyzed packet captures using Wireshark and NetworkMiner to trace malicious TCP/HTTP streams and uncover command-and-control (C2) servers.
- Extracted embedded payloads and malicious artifacts, correlating indicators of compromise (IOCs) across VirusTotal and ANY.RUN.
- Documented findings, forensic timelines, and evidence-based remediation recommendations.

TryHackMe — Hands-On Cybersecurity Learning

2025 | Platform Profile

- Completed 22+ interactive security labs covering penetration testing methodology, network enumeration, privilege escalation, and web security.
- Demonstrated consistent problem-solving in defensive and offensive techniques, achieving a ranking in the top 25% of active global learners.

National Cybersecurity Training Program — MUST

2025 | Mirpur University of Science & Technology

- Completed university-sponsored training exploring SOC operations, incident response workflows, ethical hacking concepts, and defensive security architectures.

PROFESSIONAL EXPERIENCE

Full Stack Web Developer — HexaForge

June 2025 – Present | Pakistan

Collaborative environment engineering secure web applications and robust backend services

- Developed full-stack web applications adhering to secure coding best practices, least privilege principles, and modular architecture.
- Implemented token-based authentication (JWT) and authorization guards in Python and FastAPI to safeguard backend API endpoints against unauthorized access.
- Administered PostgreSQL databases, designing relational constraints and schema protections to prevent SQL injection and ensure transactional integrity.
- Containerized application environments using Docker to enforce reproducible dependencies and isolated execution boundaries.
- Utilized Redis and Celery for background processing, decoupling heavy workloads from synchronous API threads.
- Built responsive, accessible frontend interfaces in React.js and Tailwind CSS with sanitized user inputs.

SELECTED SOFTWARE & SECURITY PROJECTS

Digital Assets Marketplace (Secure Full-Stack Platform)

React.js, FastAPI, PostgreSQL, Docker

- Engineered a decoupled digital asset platform featuring authenticated user flows and catalog management.
- Implemented robust backend API validation and parameterized database queries with FastAPI and PostgreSQL to mitigate injection attacks.
- Containerized multi-service deployment with Docker for controlled container isolation.

Student Issue Management System

Python, FastAPI, JavaScript, PostgreSQL | GitHub

- Built a university ticket-tracking system featuring structured workflow states and audit logs.
- Enforced backend authorization rules to ensure students and administrators only access permissible data.
- Structured relational records in PostgreSQL to ensure data integrity and traceability throughout the issue lifecycle.

Islamabad NGO Donation Website

PHP, MySQL, HTML5, CSS3, JavaScript | GitHub

- Developed an NGO donation portal providing transparent contribution logging and campaign tracking.
- Applied input validation and structured relational database operations in MySQL to ensure accountability.

CERTIFICATIONS & TECHNICAL LEARNING

- **Microsoft Learn Cloud & Security Achievements (2024–2025):** Completed curriculum modules covering cloud security fundamentals, governance, compliance, monitoring, and infrastructure security controls (Verification).
- **HackerRank Skill Certifications (2025):** Verified problem-solving credentials in C++ and MySQL relational database management (Verification).

EDUCATION

Bachelor of Science in Computer Science (BSCS)

Graduation: October 2026

Mirpur University of Science and Technology (MUST), Pakistan | Official DMC / transcript pending formal issuance

Relevant Coursework: Information & Network Security, Computer Networks, Operating Systems, Database Systems, Data Structures & Algorithms, Software Engineering.